

rph8-g15q: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
rph8-g15q_EQ0018 (18)	5	0.837	$\begin{aligned} &\text{\texttt{\textbackslash begin{aligned} \text{ SWAP } L_{1} \text{ SWAP } }} \\ &\text{\texttt{\textbackslash \&=L_{2} \textbackslash \& \left\{X \otimes X, L_{1}\right\} =0}} \\ &\text{\texttt{\textbackslash end{aligned}}} \end{aligned}$	$\begin{aligned} \text{SWAP } L_1 \text{ SWAP } &= L_2 \\ \{X \otimes X, L_1\} &= 0 \end{aligned}$	$\begin{aligned} \text{SWAP } L_1 \text{ SWAP } &= L_2, \\ \{X \otimes X, L_1\} &= 0 \end{aligned}$
rph8-g15q_EQ0065 (D1)	19	0.848	$\begin{aligned} &\text{\texttt{\textbackslash begin{aligned} d\left(x_{2}, x_{1}\right) \&=d y\left(f\right.}} \\ &\text{\texttt{\textbackslash \left(x_{2}\right), f\left(x_{1}\right)\right)\right\} \&=d y\left(f\left(x_{1}\right), f\left(x_{2}\right)\right)} \\ &\text{\texttt{\textbackslash \left(f\left(x_{1}\right), f\left(x_{2}\right)\right)\right)\right\} \&=d_{\mathcal{X}}\left(x_{1}, x_{2}\right)}} \\ &\text{\texttt{\textbackslash \&=d_{\mathcal{X}}\left(x_{1}, x_{2}\right) \textbackslash end{aligned}}} \end{aligned}$	$\begin{aligned} d\left(x_2, x_1\right) &= d y\left(f\left(x_2\right), f\left(x_1\right)\right) \\ &= d y\left(f\left(x_1\right), f\left(x_2\right)\right) \\ &= d_{\mathcal{X}}\left(x_1, x_2\right) \end{aligned}$	$\begin{aligned} d\left(x_2, x_1\right) &= d_{\mathcal{Y}}\left(f\left(x_2\right), f\left(x_1\right)\right) \\ &= d_{\mathcal{Y}}\left(f\left(x_1\right), f\left(x_2\right)\right) \\ &= d_{\mathcal{X}}\left(x_1, x_2\right) \end{aligned}$
rph8-g15q_EQ0061 (B12)	19	0.890	$\begin{array}{l} Z \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Z, \quad X \otimes Z, \quad Z \otimes X, \\ Y \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Y, \quad X \otimes Y, \quad Y \otimes X, \\ X, \quad Y \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes X, \quad X \otimes Y, \quad Y \otimes X, \\ Y, \quad X \otimes Y, \quad Y \otimes X, \quad X \otimes Y, \quad Y \otimes X, \end{array}$	$\begin{array}{l} Z \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Z, \quad X \otimes Z, \quad Z \otimes X, \\ Y \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Y, \quad X \otimes Y, \quad Y \otimes X, \end{array}$	$\begin{array}{l} Z \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Z, \quad X \otimes Z, \quad Z \otimes X, \\ Y \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Y, \quad X \otimes Y, \quad Y \otimes X, \end{array}$
rph8-g15q_EQ0019 (19)	5	0.921	$\begin{array}{l} Z \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Z, \quad Y \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Y, \\ X \otimes Z, \quad Z \otimes X, \quad X \otimes Y, \quad Y \otimes X, \\ X, \quad X \otimes Y, \quad Y \otimes X, \quad X \otimes Y, \quad Y \otimes X, \end{array}$	$\begin{array}{l} Z \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Z, \quad Y \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Y, \\ X \otimes Z, \quad Z \otimes X, \quad X \otimes Y, \quad Y \otimes X, \end{array}$	$\begin{array}{l} Z \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Z, \quad Y \otimes \mathbb{1}_2, \quad \mathbb{1}_2 \otimes Y, \\ X \otimes Z, \quad Z \otimes X, \quad X \otimes Y, \quad Y \otimes X, \end{array}$
rph8-g15q_EQ0049 (A16)	18	0.933	$U(x)=e^{\{-i\mathcal{L}(x)\}},$	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$
rph8-g15q_EQ0007 (7)	5	0.945	$\begin{array}{l} \alpha_{(0,0)}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \alpha_{(1,0)}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right], \\ \alpha_{(0,1)}=\left[\begin{array}{ll} -1 & 0 \\ 0 & -1 \end{array}\right], \quad \alpha_{(1,1)}=\left[\begin{array}{ll} 0 & -1 \\ -1 & 0 \end{array}\right]. \end{array}$	$\begin{array}{l} \alpha_{(0,0)}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \alpha_{(1,0)}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right], \\ \alpha_{(0,1)}=\left[\begin{array}{ll} -1 & 0 \\ 0 & -1 \end{array}\right], \quad \alpha_{(1,1)}=\left[\begin{array}{ll} 0 & -1 \\ -1 & 0 \end{array}\right]. \end{array}$	$\begin{array}{l} \alpha_{(0,0)}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad \alpha_{(1,0)}=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right], \\ \alpha_{(0,1)}=\left[\begin{array}{ll} -1 & 0 \\ 0 & -1 \end{array}\right], \quad \alpha_{(1,1)}=\left[\begin{array}{ll} 0 & -1 \\ -1 & 0 \end{array}\right]. \end{array}$
rph8-g15q_EQ0064 (B9)	19	0.951	$\begin{aligned} &\text{\texttt{\textbackslash begin{aligned} d_{\mathcal{A}}\left(x_{1}, x_{2}\right) \&=\sqrt{\left\{ \right.}}}} \\ &\text{\texttt{\textbackslash \left(x_{1}-x_{2}\right)^{\mathcal{T}} V^{\mathcal{T}} B V\left(x_{1}-x_{2}\right)}} \\ &\text{\texttt{\textbackslash \left(V x_{1}-V x_{2}\right)^{\mathcal{T}} B \left(V x_{1}-V x_{2}\right)\right\} \&=}} \\ &\text{\texttt{\textbackslash \left(V x_{1}-V x_{2}\right)^{\mathcal{T}} B \left(V x_{1}-V x_{2}\right)\right\} \&=d_{\mathcal{B}}\left(V x_{1}, V x_{2}\right) \textbackslash end{aligned}}} \end{aligned}$	$\begin{aligned} d_{\mathcal{A}}\left(x_1, x_2\right) &= \sqrt{\left(x_1-x_2\right)^T V^T B V\left(x_1-x_2\right)} \\ &= \sqrt{\left(V x_1-V x_2\right)^T B\left(V x_1-V x_2\right)} \\ &= d_{\mathcal{B}}\left(V x_1, V x_2\right) \end{aligned}$	$\begin{aligned} d_{\mathcal{A}}\left(x_1, x_2\right) &= \sqrt{\left(x_1-x_2\right)^T V^T B V\left(x_1-x_2\right)} \\ &= \sqrt{\left(V x_1-V x_2\right)^T B\left(V x_1-V x_2\right)} \\ &= d_{\mathcal{B}}\left(V x_1, V x_2\right), \end{aligned}$
rph8-g15q_EQ0008 (8)	5	0.962	$\begin{array}{l} V_{(0,0)}=\mathbb{1}_2 \otimes \mathbb{1}_2, \quad V_{(1,0)}=\text{operatorname{SWAP}}, \\ V_{(0,1)}=X \otimes X, \quad V_{(1,1)}=\text{operatorname{SWAP}}(X \otimes X), \end{array}$	$\begin{array}{l} V_{(0,0)}=\mathbb{1}_2 \otimes \mathbb{1}_2, \quad V_{(1,0)}=\text{SWAP}, \\ V_{(0,1)}=X \otimes X, \quad V_{(1,1)}=\text{SWAP}(X \otimes X), \end{array}$	$\begin{array}{l} V_{(0,0)}=\mathbb{1}_2 \otimes \mathbb{1}_2, \quad V_{(1,0)}=\text{SWAP}, \\ V_{(0,1)}=X \otimes X, \quad V_{(1,1)}=\text{SWAP}(X \otimes X), \end{array}$

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rph8-g15q_ EQ0046 (A13)	17	■0.985	\begin{aligned} y &= \left.\frac{d}{dt} \exp (t y)\right _{t=0}=\left.\frac{d}{d t}\left\{f \circ \gamma\right\}\right _{t=0} \\ &\quad \text { by Eq. (A12) } \\ &= \left.D f\right _{1_G}\left.\left(\frac{d}{d t} \gamma\right)\right _{t=0} \\ &\quad \text { by the chain rule } \\ &= \left.D f\right _{1_G}\left.\left(\frac{d}{d t} \exp (t x)\right)\right _{t=0} \\ &\quad \text { by Eq. (A11) } \\ &= \mathcal{M}(x) \quad \text { by definition of } \mathcal{M} \end{aligned}	$y=\left.\frac{d}{d t} \exp (t y)\right _{t=0}=\left.\frac{d}{d t}(f \circ \gamma)\right _{t=0} \quad \text { by Eq. (A12) }$ $=D f _{1_G}\left(\left.\frac{d}{d t} \gamma\right _{t=0}\right) \quad \text { by the chain rule }$ $=D f _{1_G}\left(\left.\frac{d}{d t} \exp (t x)\right _{t=0}\right) \quad \text { by Eq. (A11) }$ $=\mathcal{M}(x) \quad \text { by definition of } \mathcal{M}$	$y=\frac{d}{d t} \exp (t y) _{t=0}=\frac{d}{d t}(f \circ \gamma) _{t=0} \quad \text { by Eq. (A12) }$ $=D f _{1_G}\left(\frac{d}{d t} \gamma _{t=0}\right) \quad \text { by the chain rule }$ $=D f _{1_G}\left(\frac{d}{d t} \exp (t x) _{t=0}\right) \quad \text { by Eq. (A11) }$ $=\mathcal{M}(x) \quad \text { by definition of } \mathcal{M}.$ (A
rph8-g15q_ EQ0067 (D3)	19	■0.991	\begin{aligned} d_{\{\mathrm{X}\}}\left(x_1, x_2\right) &\geq d_Y\left(f\left(x_1\right), f\left(x_3\right)\right)+d_Y\left(f\left(x_3\right), f\left(x_2\right)\right) \\ &=d_{\mathcal{X}}\left(x_1, x_3\right)+d_{\mathcal{X}}\left(x_3, x_2\right), \end{aligned}	$d_{\mathcal{X}}\left(x_1, x_2\right) \geq d_Y\left(f\left(x_1\right), f\left(x_3\right)\right)+d_Y\left(f\left(x_3\right), f\left(x_2\right)\right)$ $=d_{\mathcal{X}}\left(x_1, x_3\right)+d_{\mathcal{X}}\left(x_3, x_2\right)$	$d_{\mathcal{X}}\left(x_1, x_2\right) \geq d_Y\left(f\left(x_1\right), f\left(x_3\right)\right)+d_Y\left(f\left(x_3\right), f\left(x_2\right)\right)$ $=d_{\mathcal{X}}\left(x_1, x_3\right)+d_{\mathcal{X}}\left(x_3, x_2\right),$
rph8-g15q_ EQ0027 (27)	10	■0.994	d_{\{\mathrm{Tr}\}}(\rho, \sigma)=\left\ \rho-\sigma\right\ _1 \equiv \mathrm{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right] .	$d_{\mathrm{Tr}}(\rho, \sigma)=\ \rho-\sigma\ _1 \equiv \mathrm{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right] .$	$d_{\mathrm{Tr}}(\rho, \sigma)=\ \rho-\sigma\ _1 \equiv \mathrm{Tr}\left[\sqrt{(\rho-\sigma)^{\dagger}(\rho-\sigma)}\right] .$
rph8-g15q_ EQ0054 (B5)	18	■0.994	\left \psi_0\right\rangle\left \right\rangle \text {rangle}=\sqrt{p} +,+\rangle-\sqrt{1-p} -, -\rangle	$ \psi_0\rangle=\sqrt{p} +,+\rangle-\sqrt{1-p} -, -\rangle$	$ \psi_0\rangle=\sqrt{p} +,+\rangle-\sqrt{1-p} -, -\rangle$
rph8-g15q_EQ0009 (9)	5	■0.996	U\left(x_1, x_2\right)=R_Z\left(x_1\right) \otimes R_Z\left(x_2\right),	$U\left(x_1, x_2\right)=R_Z\left(x_1\right) \otimes R_Z\left(x_2\right),$	$U\left(x_1, x_2\right)=R_Z\left(x_1\right) \otimes R_Z\left(x_2\right),$
rph8-g15q_ EQ0011 (11)	5	■0.999	\left \psi_0\right\rangle\left \right\rangle \text {rangle}=\sqrt{p} +,+\rangle-\sqrt{1-p} -, -\rangle,	$ \psi_0\rangle=\sqrt{p} +,+\rangle-\sqrt{1-p} -, -\rangle,$	$ \psi_0\rangle=\sqrt{p} +,+\rangle-\sqrt{1-p} -, -\rangle,$
rph8-g15q_EQ0005 (5)	5	■1.000	y\left(x_1, x_2\right)=y\left(x_2, x_1\right)=y\left(-x_1,-x_2\right),	$y\left(x_1, x_2\right)=y\left(x_2, x_1\right)=y\left(-x_1,-x_2\right),$	$y\left(x_1, x_2\right)=y\left(x_2, x_1\right)=y\left(-x_1,-x_2\right),$
rph8-g15q_ EQ0016 (16)	5	■1.000	V_g e^{-i \mathcal{L}(x)} V_g^{\dagger}=e^{-i \mathcal{L}\left(\alpha_g(x)\right)}	$V_g e^{-i \mathcal{L}(x)} V_g^{\dagger}=e^{-i \mathcal{L}\left(\alpha_g(x)\right)}$	$V_g e^{-i \mathcal{L}(x)} V_g^{\dagger}=e^{-i \mathcal{L}\left(\alpha_g(x)\right)}$
rph8-g15q_ EQ0023 (23)	7	■1.000	x \mapsto \chi(x)=\sum_{n=0}^{2^d-1} x_n n\rangle \mapsto \Phi(x)\rangle=\frac{\chi(x)}{\ \chi(x)\ },	$x \mapsto \chi(x)=\sum_{n=0}^{2^d-1} x_n n\rangle \mapsto \Phi(x)\rangle=\frac{\chi(x)}{\ \chi(x)\ },$	$x \mapsto \chi(x)=\sum_{n=0}^{2^d-1} x_n n\rangle \mapsto \Phi(x)\rangle=\frac{\chi(x)}{\ \chi(x)\ },$
rph8-g15q_ EQ0029 (29)	10	■1.000	d_B(\rho, \sigma)=\sqrt{2\left(1-\sqrt{F(\rho, \sigma)}\right)},	$d_B(\rho, \sigma)=\sqrt{2\left(1-\sqrt{F(\rho, \sigma)}\right)},$	$d_B(\rho, \sigma)=\sqrt{2\left(1-\sqrt{F(\rho, \sigma)}\right)},$
rph8-g15q_ EQ0030 (30)	10	■1.000	F(\rho, \sigma)=\left(\mathrm{Tr}\left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}\right]\right)^2	$F(\rho, \sigma)=\left(\mathrm{Tr}\left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}\right]\right)^2$	$F(\rho, \sigma)=\left(\mathrm{Tr}\left[\sqrt{\sqrt{\rho} \sigma \sqrt{\rho}}\right]\right)^2$
rph8-g15q_ EQ0032 (32)	13	■1.000	F\left(\mathrm{id}_{(\mathcal{X}, \alpha)}\right)=\mathrm{id}_{(\mathcal{X}, \alpha)}=\mathrm{id}_{\mathcal{X}}=\mathrm{id}_{F(\mathcal{X}, \alpha)} .	$F\left(\mathrm{id}_{(\mathcal{X}, \alpha)}\right)=\mathrm{id}_{(\mathcal{X}, \alpha)}=\mathrm{id}_{\mathcal{X}}=\mathrm{id}_{F(\mathcal{X}, \alpha)} .$	$F\left(\mathrm{id}_{(\mathcal{X}, \alpha)}\right)=\mathrm{id}_{(\mathcal{X}, \alpha)}=\mathrm{id}_{\mathcal{X}}=\mathrm{id}_{F(\mathcal{X}, \alpha)} .$
rph8-g15q_EQ0001 (1)	3	■1.000	\rho(x)=U(x) \psi_0\rangle\langle\psi_0 U(x)^{\dagger} .	$\rho(x)=U(x) \psi_0\rangle\langle\psi_0 U(x)^{\dagger} .$	$\rho(x)=U(x) \psi_0\rangle\langle\psi_0 U(x)^{\dagger} .$
rph8-g15q_EQ0002 (2)	4	■1.000	\operatorname{Ad}_{V_g}(\sigma):=V_g \sigma V_g^{\dagger}	$\operatorname{Ad}_{V_g}(\sigma):=V_g \sigma V_g^{\dagger}$	$\operatorname{Ad}_{V_g}(\sigma):=V_g \sigma V_g^{\dagger}$

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rph8-g15q_EQ0003 (3)	4	■ 1.000	$\rho\left(\alpha_g(x)\right)=V_{\{g\}}\rho(x)V_{\{g\}}^{\dagger}$	$\rho\left(\alpha_g(x)\right)=V_g\rho(x)V_g^{\dagger}$	$\rho(\alpha_g(x))=V_g\rho(x)V_g^{\dagger}$
rph8-g15q_EQ0004 (4)	4	■ 1.000	$f\left(\alpha_g(x)\right)=\beta_g(f(x))$	$f\left(\alpha_g(x)\right)=\beta_g(f(x))$	$f(\alpha_g(x))=\beta_g(f(x))$
rph8-g15q_EQ0006 (6)	5	■ 1.000	$G=\mathbb{Z}_2\times\mathbb{Z}_2=\{(0,0),(1,0),(0,1),(1,1)\}$	$G=\mathbb{Z}_2\times\mathbb{Z}_2=\{(0,0),(1,0),(0,1),(1,1)\}$	$G=\mathbb{Z}_2\times\mathbb{Z}_2=\{(0,0),(1,0),(0,1),(1,1)\}$
rph8-g15q_EQ0010 (10)	5	■ 1.000	$R_Z(\theta)=e^{-i\theta Z/2}=\left[\begin{array}{cc}e^{-i\theta/2}&0\\0&e^{i\theta/2}\end{array}\right]$	$R_Z(\theta)=e^{-i\theta Z/2}=\left[\begin{array}{cc}e^{-i\theta/2}&0\\0&e^{i\theta/2}\end{array}\right]$	$R_Z(\theta)=e^{-i\theta Z/2}=\left[\begin{array}{cc}e^{-i\theta/2}&0\\0&e^{i\theta/2}\end{array}\right],$
rph8-g15q_EQ0012 (12)	5	■ 1.000	$ +\rangle=\frac{1}{\sqrt{2}}(0\rangle+ 1\rangle)\quad\text{and}\quad -\rangle=\frac{1}{\sqrt{2}}(0\rangle- 1\rangle)$	$ +\rangle=\frac{1}{\sqrt{2}}(0\rangle+ 1\rangle)\quad\text{and}\quad -\rangle=\frac{1}{\sqrt{2}}(0\rangle- 1\rangle)$	$ +\rangle=\frac{1}{\sqrt{2}}(0\rangle+ 1\rangle)\quad\text{and}\quad -\rangle=\frac{1}{\sqrt{2}}(0\rangle- 1\rangle)$
rph8-g15q_EQ0013 (13)	5	■ 1.000	$\rho(x)=U(x_1,x_2) \psi_0\rangle\langle\psi_0 U(x_1,x_2)^{\dagger}.$	$\rho(x)=U(x_1,x_2) \psi_0\rangle\langle\psi_0 U(x_1,x_2)^{\dagger}.$	$\rho(x)=U(x_1,x_2) \psi_0\rangle\langle\psi_0 U(x_1,x_2)^{\dagger}.$
rph8-g15q_EQ0014 (14)	5	■ 1.000	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$
rph8-g15q_EQ0015 (15)	5	■ 1.000	$L_1:=\mathcal{L}(e_1)\quad\text{and}\quad L_2:=\mathcal{L}(e_2),$	$L_1:=\mathcal{L}(e_1)\quad\text{and}\quad L_2:=\mathcal{L}(e_2),$	$L_1:=\mathcal{L}(e_1)\quad\text{and}\quad L_2:=\mathcal{L}(e_2),$
rph8-g15q_EQ0017 (17)	5	■ 1.000	$V_{\{g\}}\mathcal{L}(x)V_{\{g\}}^{\dagger}=\mathcal{L}(\alpha_g(x))$	$V_g\mathcal{L}(x)V_g^{\dagger}=\mathcal{L}(\alpha_g(x))$	$V_g\mathcal{L}(x)V_g^{\dagger}=\mathcal{L}(\alpha_g(x))$
rph8-g15q_EQ0020 (20)	6	■ 1.000	$L_1=\frac{Z}{2}\otimes\mathbb{1}_2\quad\text{and}\quad L_2=\mathbb{1}_2\otimes\frac{Z}{2}$	$L_1=\frac{Z}{2}\otimes\mathbb{1}_2\quad\text{and}\quad L_2=\mathbb{1}_2\otimes\frac{Z}{2}$	$L_1=\frac{Z}{2}\otimes\mathbb{1}_2\quad\text{and}\quad L_2=\mathbb{1}_2\otimes\frac{Z}{2}$
rph8-g15q_EQ0021 (21)	6	■ 1.000	$x\mapsto \Phi(x)\rangle:=\left(\bigotimes_{k=1}^d\exp\left(-\frac{ix_k}{2}X_k\right)\right) 0\cdots 0\rangle,$	$x\mapsto \Phi(x)\rangle:=\left(\bigotimes_{k=1}^d\exp\left(-\frac{ix_k}{2}X_k\right)\right) 0\cdots 0\rangle,$	$x\mapsto \Phi(x)\rangle:=\left(\bigotimes_{k=1}^d\exp\left(-\frac{ix_k}{2}X_k\right)\right) 0\cdots 0\rangle,$
rph8-g15q_EQ0022 (22)	6	■ 1.000	$\rho(x):= \Phi(x)\rangle\langle\Phi(x) .$	$\rho(x):= \Phi(x)\rangle\langle\Phi(x) .$	$\rho(x):= \Phi(x)\rangle\langle\Phi(x) .$
rph8-g15q_EQ0024 (24)	7	■ 1.000	$(x_0,x_1,\dots,x_{N-1})\mapsto\frac{(x_0,x_1,\dots,x_{N-1},1)}{\sqrt{1+\ x\ ^2}},$	$(x_0,x_1,\dots,x_{N-1})\mapsto\frac{(x_0,x_1,\dots,x_{N-1},1)}{\sqrt{1+\ x\ ^2}},$	$(x_0,x_1,\dots,x_{N-1})\mapsto\frac{(x_0,x_1,\dots,x_{N-1},1)}{\sqrt{1+\ x\ ^2}},$
rph8-g15q_EQ0025 (25)	7	■ 1.000	$(x_0,\dots,x_{N-1})\mapsto\frac{(e^{x_0},\dots,e^{x_{N-1}},1)}{1+\sum_{j=0}^{N-1}e^{x_j}},$	$(x_0,\dots,x_{N-1})\mapsto\frac{(e^{x_0},\dots,e^{x_{N-1}},1)}{1+\sum_{j=0}^{N-1}e^{x_j}},$	$(x_0,\dots,x_{N-1})\mapsto\frac{(e^{x_0},\dots,e^{x_{N-1}},1)}{1+\sum_{j=0}^{N-1}e^{x_j}},$
rph8-g15q_EQ0026 (26)	9	■ 1.000	$d_{\mathcal{X}}(x_1,x_2):=d_{\mathcal{Y}}(f(x_1),f(x_2))$	$d_{\mathcal{X}}(x_1,x_2):=d_{\mathcal{Y}}(f(x_1),f(x_2))$	$d_{\mathcal{X}}(x_1,x_2):=d_{\mathcal{Y}}(f(x_1),f(x_2))$
rph8-g15q_EQ0028 (28)	10	■ 1.000	$d_{\mathrm{HS}}(\rho,\sigma)=\sqrt{\mathrm{Tr}[(\rho-\sigma)^{\dagger}(\rho-\sigma)]}.$	$d_{\mathrm{HS}}(\rho,\sigma)=\sqrt{\mathrm{Tr}[(\rho-\sigma)^{\dagger}(\rho-\sigma)]}.$	$d_{\mathrm{HS}}(\rho,\sigma)=\sqrt{\mathrm{Tr}[(\rho-\sigma)^{\dagger}(\rho-\sigma)]}.$
rph8-g15q_EQ0031 (31)	13	■ 1.000	$F(g\circ f)=g\circ f=F(g)\circ F(f).$	$F(g\circ f)=g\circ f=F(g)\circ F(f).$	$F(g\circ f)=g\circ f=F(g)\circ F(f).$
rph8-g15q_EQ0033 (33)	14	■ 1.000	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$
rph8-g15q_EQ0034 (A1)	16	■ 1.000	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$	$U(x)=e^{-i\mathcal{L}(x)},$
rph8-g15q_EQ0035 (A2)	16	■ 1.000	$\gamma(t)=\exp(tM)$	$\gamma(t)=\exp(tM)$	$\gamma(t)=\exp(tM)$

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rph8-g15q_ EQ0036 (A3)	16	■ 1.000	$f(\exp(W))=\exp(\mathcal{M}(W))$	$f(\exp(W)) = \exp(\mathcal{M}(W))$	$f(\exp(W)) = \exp(\mathcal{M}(W))$
rph8-g15q_ EQ0037 (A4)	16	■ 1.000	$\left[\mathcal{M}(W_1), \mathcal{M}(W_2)\right] = \mathcal{M}([W_1, W_2])$	$[\mathcal{M}(W_1), \mathcal{M}(W_2)] = \mathcal{M}([W_1, W_2])$	$[\mathcal{M}(W_1), \mathcal{M}(W_2)] = \mathcal{M}([W_1, W_2])$
rph8-g15q_ EQ0038 (A5)	17	■ 1.000	$U((s, t))=e^{-i s X-i t Y}$	$U((s, t)) = e^{-isX-itY}$	$U((s, t)) = e^{-isX-itY}$
rph8-g15q_ EQ0039 (A6)	17	■ 1.000	$\begin{aligned} U((s, 0)) U((0, t)) &= e^{-i s X} e^{-i t Y} \\ &\neq e^{-i(s X+t Y)} \\ &= U((s, 0)+(0, t)) \end{aligned}$	$\begin{aligned} U((s, 0)) U((0, t)) &= e^{-isX} e^{-itY} \\ &\neq e^{-i(sX+tY)} \\ &= U((s, 0) + (0, t)) \end{aligned}$	$\begin{aligned} U((s, 0)) U((0, t)) &= e^{-isX} e^{-itY} \\ &\neq e^{-i(sX+tY)} \\ &= U((s, 0) + (0, t)) \end{aligned}$
rph8-g15q_ EQ0040 (A7)	17	■ 1.000	$\mathbb{R}_{\alpha}=\left\{r \alpha: \alpha \in \mathbb{R}^2 \backslash\{0\}, r \in \mathbb{R}\right\},$	$\mathbb{R}_{\alpha}=\left\{r \alpha: \alpha \in \mathbb{R}^2 \backslash\{0\}, r \in \mathbb{R}\right\},$	$\mathbb{R}_{\alpha}=\left\{r \alpha: \alpha \in \mathbb{R}^2 \backslash\{0\}, r \in \mathbb{R}\right\},$
rph8-g15q_ EQ0041 (A8)	17	■ 1.000	$\begin{aligned} U(r \alpha) U(u \alpha) &= e^{-i r W} e^{-i u W} \\ &= e^{-i(r+u) W} \\ &= U(r \alpha+u \alpha), \end{aligned}$	$\begin{aligned} U(r \alpha) U(u \alpha) &= e^{-i r W} e^{-i u W} \\ &= e^{-i(r+u) W} \\ &= U(r \alpha+u \alpha), \end{aligned}$	$\begin{aligned} U(r \alpha) U(u \alpha) &= e^{-i r W} e^{-i u W} \\ &= e^{-i(r+u) W} \\ &= U(r \alpha+u \alpha), \end{aligned}$
rph8-g15q_ EQ0042 (A9)	17	■ 1.000	$\mathbb{R} \stackrel{\alpha}{\rightarrow} \mathbb{R}^2 \stackrel{U}{\rightarrow} \mathcal{U}\left(\mathbb{C}^2\right)$	$\mathbb{R} \stackrel{\alpha}{\rightarrow} \mathbb{R}^2 \stackrel{U}{\rightarrow} \mathcal{U}\left(\mathbb{C}^2\right)$	$\mathbb{R} \stackrel{\alpha}{\rightarrow} \mathbb{R}^2 \stackrel{U}{\rightarrow} \mathcal{U}\left(\mathbb{C}^2\right)$
rph8-g15q_ EQ0043 (A10)	17	■ 1.000	$f(\exp(x))=\exp(\mathcal{M}(x))$	$f(\exp(x)) = \exp(\mathcal{M}(x))$	$f(\exp(x)) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0044 (A11)	17	■ 1.000	$\gamma(t)=\exp (t x)$	$\gamma(t) = \exp(tx)$	$\gamma(t) = \exp(tx)$
rph8-g15q_ EQ0045 (A12)	17	■ 1.000	$(f \circ \gamma)(t)=\exp (t y)$	$(f \circ \gamma)(t) = \exp(ty)$	$(f \circ \gamma)(t) = \exp(ty)$
rph8-g15q_ EQ0047 (A14)	17	■ 1.000	$f(\exp (t x))=\exp (t \mathcal{M}(x))$	$f(\exp(tx)) = \exp(t\mathcal{M}(x))$	$f(\exp(tx)) = \exp(t\mathcal{M}(x))$
rph8-g15q_ EQ0048 (A15)	18	■ 1.000	$f(x)=\exp (\mathcal{M}(x))$	$f(x) = \exp(\mathcal{M}(x))$	$f(x) = \exp(\mathcal{M}(x))$
rph8-g15q_ EQ0050 (B1)	18	■ 1.000	$c\left(x_1, x_2\right)=c\left(x_2, x_1\right)=c\left(-x_1,-x_2\right) .$	$c\left(x_1, x_2\right)=c\left(x_2, x_1\right)=c\left(-x_1,-x_2\right) .$	$c\left(x_1, x_2\right)=c\left(x_2, x_1\right)=c\left(-x_1,-x_2\right) .$
rph8-g15q_ EQ0051 (B2)	18	■ 1.000	$c(x)=\begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}$	$c(x)=\begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}$	$c(x)=\begin{cases}-1, & \text { if } y(x)<0, \\ 0, & \text { if } y(x)=0, \\ +1, & \text { if } y(x)>0, \end{cases}$
rph8-g15q_ EQ0052 (B3)	18	■ 1.000	$y(x)=\operatorname{Tr}[\rho(x) O],$	$y(x) = \operatorname{Tr}[\rho(x)O],$	$y(x) = \operatorname{Tr}[\rho(x)O],$
rph8-g15q_ EQ0053 (B4)	18	■ 1.000	$y\left(x_1, x_2\right)=y\left(x_2, x_1\right)=y\left(-x_1,-x_2\right) .$	$y\left(x_1, x_2\right)=y\left(x_2, x_1\right)=y\left(-x_1,-x_2\right) .$	$y\left(x_1, x_2\right)=y\left(x_2, x_1\right)=y\left(-x_1,-x_2\right) .$
rph8-g15q_ EQ0055 (B6)	18	■ 1.000	$O=X \otimes X,$	$O = X \otimes X,$	$O = X \otimes X,$
rph8-g15q_ EQ0056 (B7)	18	■ 1.000	$y\left(x_1, x_2\right)=\cos \left(x_1\right) \sin \left(x_2\right)+2 \sqrt{p(1-p)} \sin \left(x_1\right) \sin \left(x_2\right) .$	$y\left(x_1, x_2\right)=\cos \left(x_1\right) \sin \left(x_2\right)+2 \sqrt{p(1-p)} \sin \left(x_1\right) \sin \left(x_2\right) .$	$y\left(x_1, x_2\right)=\cos \left(x_1\right) \sin \left(x_2\right)+2 \sqrt{p(1-p)} \sin \left(x_1\right) \sin \left(x_2\right) .$

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rph8-g15q_ EQ0057 (B8)	18	■1.000	$L_1=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ & \overline{a_{12}} & & \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array}\right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array} \right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \end{array} \right],$
rph8-g15q_ EQ0058 (C3)	19	■1.000	$\left[\begin{array}{cccc} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array}\right] = - \left[\begin{array}{cccc} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array}\right].$	$\left[\begin{array}{cccc} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array} \right] = - \left[\begin{array}{cccc} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array} \right].$	$\left[\begin{array}{cccc} \overline{a_{14}} & \overline{a_{24}} & \overline{a_{34}} & a_{44} \\ \overline{a_{13}} & \overline{a_{23}} & a_{33} & a_{34} \\ \overline{a_{12}} & a_{22} & a_{23} & a_{24} \\ a_{11} & a_{12} & a_{13} & a_{14} \end{array} \right] = - \left[\begin{array}{cccc} a_{14} & a_{13} & a_{12} & a_{11} \\ a_{24} & a_{23} & a_{22} & \overline{a_{12}} \\ a_{34} & a_{33} & \overline{a_{23}} & \overline{a_{13}} \\ a_{44} & \overline{a_{34}} & \overline{a_{24}} & \overline{a_{14}} \end{array} \right].$
rph8-g15q_ EQ0059 (B10)	19	■1.000	$L_1=\left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ & \overline{a_{12}} & & \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array}\right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array} \right],$	$L_1 = \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & a_{14} \\ \overline{a_{12}} & a_{22} & a_{23} & -\overline{a_{13}} \\ \overline{a_{13}} & -a_{23} & -a_{22} & -\overline{a_{12}} \\ -a_{14} & -a_{13} & -a_{12} & -a_{11} \end{array} \right],$
rph8-g15q_ EQ0060 (B11)	19	■1.000	$a_{12}, a_{13} \in \mathbb{C}, \quad \text{\texttt{\textbackslash quad}} \quad a_{14}, a_{23} \in i\mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R}.$	$a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i\mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R}.$	$a_{12}, a_{13} \in \mathbb{C}, \quad a_{14}, a_{23} \in i\mathbb{R}, \quad a_{11}, a_{22} \in \mathbb{R}.$
rph8-g15q_ EQ0062 (C1)	19	■1.000	$d_{\mathcal{X}}(x_1, x_2) = \sqrt{(x_1 - x_2)^T (V^T V) (x_1 - x_2)},$	$d_{\mathcal{X}}(x_1, x_2) = \sqrt{(x_1 - x_2)^T (V^T V) (x_1 - x_2)},$	$d_{\mathcal{X}}(x_1, x_2) = \sqrt{(x_1 - x_2)^T (V^T V) (x_1 - x_2)},$
rph8-g15q_ EQ0063 (C2)	19	■1.000	$d_A(x_1, x_2) := \sqrt{(x_1 - x_2)^T A (x_1 - x_2)}.$	$d_A(x_1, x_2) := \sqrt{(x_1 - x_2)^T A (x_1 - x_2)}.$	$d_A(x_1, x_2) := \sqrt{(x_1 - x_2)^T A (x_1 - x_2)}.$
rph8-g15q_ EQ0066 (D2)	19	■1.000	$\begin{aligned} & d_{\mathcal{X}}(x_1, x_2) = d_Y(f(x_1), f(x_2)) \\ & \geq d_Y(f(x_1), y) + d_Y(y, f(x_2)) \end{aligned}$	$\begin{aligned} d_{\mathcal{X}}(x_1, x_2) &= d_Y(f(x_1), f(x_2)) \\ &\geq d_Y(f(x_1), y) + d_Y(y, f(x_2)) \end{aligned}$	$\begin{aligned} d_{\mathcal{X}}(x_1, x_2) &= d_Y(f(x_1), f(x_2)) \\ &\geq d_Y(f(x_1), y) + d_Y(y, f(x_2)) \end{aligned}$
rph8-g15q_ EQ0068 (D4)	19	■1.000	$0 = d_{\mathcal{X}}(x_1, x_2) = d_Y(f(x_1), f(x_2)),$	$0 = d_{\mathcal{X}}(x_1, x_2) = d_Y(f(x_1), f(x_2)),$	$0 = d_{\mathcal{X}}(x_1, x_2) = d_Y(f(x_1), f(x_2)),$
rph8-g15q_ EQ0069 (D5)	20	■1.000	$c(x) = \begin{cases} +1, & \text{if } x \in A, \\ -1, & \text{if } x \in B. \end{cases}$	$c(x) = \begin{cases} +1, & \text{if } x \in A, \\ -1, & \text{if } x \in B. \end{cases}$	$c(x) = \begin{cases} +1, & \text{if } x \in A, \\ -1, & \text{if } x \in B. \end{cases}$
rph8-g15q_ EQ0070 (D6)	20	■1.000	$\rho_A := \frac{1}{\#A} \sum_{a \in A} \rho(a) \quad \text{\texttt{\textbackslash quad}} \quad \rho_B := \frac{1}{\#B} \sum_{b \in B} \rho(b),$	$\rho_A := \frac{1}{\#A} \sum_{a \in A} \rho(a) \quad \text{and} \quad \rho_B := \frac{1}{\#B} \sum_{b \in B} \rho(b),$	$\rho_A := \frac{1}{\#A} \sum_{a \in A} \rho(a) \quad \text{and} \quad \rho_B := \frac{1}{\#B} \sum_{b \in B} \rho(b),$

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rph8-g15q_ EQ0071 (D7)	20	■ 1.000	$C\left(\rho_A, \rho_B\right)=1-\frac{1}{2} d_{\mathrm{HS}}\left(\rho_A, \rho_B\right) .$	$C\left(\rho_A, \rho_B\right)=1-\frac{1}{2} d_{\mathrm{HS}}\left(\rho_A, \rho_B\right) .$	$C\left(\rho_A, \rho_B\right)=1-\frac{1}{2} d_{\mathrm{HS}}\left(\rho_A, \rho_B\right) .$
rph8-g15q_ EQ0072 (D8)	20	■ 1.000	$U_{\theta}(x)=\left(\prod_{n=1}^4 R_X(x) R_Y\left(\theta_n\right)\right) R_X(x),$	$U_{\theta}(x)=\left(\prod_{n=1}^4 R_X(x) R_Y\left(\theta_n\right)\right) R_X(x),$	$U_{\theta}(x)=\left(\prod_{n=1}^4 R_X(x) R_Y\left(\theta_n\right)\right) R_X(x),$
rph8-g15q_ EQ0073 (D9)	20	■ 1.000	$\prod_{n=1}^N A_n=A_N \cdots A_1 .$	$\prod_{n=1}^N A_n=A_N \cdots A_1 .$	$\prod_{n=1}^N A_n=A_N \cdots A_1 .$
rph8-g15q_ EQ0074 (D10)	20	■ 1.000	$\rho_{\theta}(x)= x\rangle\langle x ,$	$\rho_{\theta}(x)= x\rangle\langle x ,$	$\rho_{\theta}(x)= x\rangle\langle x ,$
rph8-g15q_ EQ0075 (D11)	20	■ 1.000	$ x\rangle=U_{\theta}(x) 0\rangle .$	$ x\rangle=U_{\theta}(x) 0\rangle .$	$ x\rangle=U_{\theta}(x) 0\rangle .$
rph8-g15q_ EQ0076 (D12)	20	■ 1.000	$O:=\rho_A-\rho_B,$	$O:=\rho_A-\rho_B,$	$O:=\rho_A-\rho_B,$
rph8-g15q_ EQ0077 (D13)	20	■ 1.000	$y(x)=\operatorname{Tr}\left[\rho(x) O\right] .$	$y(x)=\operatorname{Tr}\left[\rho(x) O\right] .$	$y(x)=\operatorname{Tr}\left[\rho(x) O\right] .$
rph8-g15q_ EQ0078 (D14)	20	■ 1.000	$\operatorname{Tr}\left[\rho_A-\rho_B\right]=\operatorname{Tr}\left[\rho_A\right]-\operatorname{Tr}\left[\rho_B\right]=1-1=0,$	$\operatorname{Tr}\left[\rho_A-\rho_B\right]=\operatorname{Tr}\left[\rho_A\right]-\operatorname{Tr}\left[\rho_B\right]=1-1=0,$	$\operatorname{Tr}\left[\rho_A-\rho_B\right]=\operatorname{Tr}\left[\rho_A\right]-\operatorname{Tr}\left[\rho_B\right]=1-1=0,$
rph8-g15q_ EQ0079 (D15)	20	■ 1.000	$\operatorname{sgn}(y(x))=c(x)$	$\operatorname{sgn}(y(x))=c(x)$	$\operatorname{sgn}(y(x))=c(x)$
rph8-g15q_ EQ0080 (E2)	21	■ 1.000	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text { times }} \cong \mathbb{C}^{2^n},$	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text { times }} \cong \mathbb{C}^{2^n},$	$\overbrace{\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2}^{n \text { times }} \cong \mathbb{C}^{2^n},$
rph8-g15q_ EQ0081 (E3)	21	■ 1.000	$\left x_0\right\rangle \otimes\left x_1\right\rangle \otimes \cdots \otimes\left x_{n-1}\right\rangle .$	$\left x_0 x_1 \cdots x_{n-1}\right\rangle=\left x_0\right\rangle \otimes\left x_1\right\rangle \otimes \cdots \otimes\left x_{n-1}\right\rangle .$	$\left x_0 x_1 \cdots x_{n-1}\right\rangle=\left x_0\right\rangle \otimes\left x_1\right\rangle \otimes \cdots \otimes\left x_{n-1}\right\rangle .$